STAT 7650: Computational Statistics

5. EM Optimization Methods

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Outline

1. EM algorithm
2. Variance estimation
3. Improving the E-step
4. Improving the M-step
5. Acceleration methods

Reference: Givens and Hoeting (2012). Chapter 4.

Missing Data and Latent Variables

Some applications with missing data or latent variables.

- (Multivariate regression with missing data) Let the observations be $\{(y_i, x_i), y_i \in \mathbb{R}^r, x_i \in \mathbb{R}^p, i = 1, \ldots, n\}$. It is common in practice that some elements of some y_i are missing.
- (Gaussian mixture models) Let $x_1, \ldots, x_n$ be an iid sample from a mixture of K normal distributions.

$$x_i \sim \sum_{k=1}^{K} \pi_k N(\mu_k, \sigma_k^2)$$

- (Censored data) Let $y_1, \ldots, y_n \sim \text{exponential}(\lambda)$. Some cases are right-censored and the observed data is $x_i = (\min(y_i, c_i), \delta_i)$, where c_i is the censoring level and $\delta_i = I(y_i \leq c_i)$.

Example: Peppered Moths

The coloring of peppered moth, *Biston betularia*, is believed to be determined by a single gene with three alleles, C, I, T, which yield 6 genotypes and 3 phenotypes.

Genotype	Phenotype	Description
CC, CI, CT	carbonaria	solid black coloring
TT	typica	light-colored patterned wings
II, IT	insularia	mottled with intermediate color



Example: Multinomial Distribution

Capture n moths and classify them according to their phenotypes.

- observed data $x = (n_C, n_I, n_T)$ with sum n
- complete data $y = (n_{CC}, n_{CI}, n_{CT}, n_{II}, n_{IT}, n_{TT})$ with sum n

Let the allele probabilities be p_C, p_I, p_T with $p_C + p_I + p_T = 1$. The goal is to estimate p_C, p_I, p_T.

Phenotype	carbonaria			insularia		typica	total
observed data x	n_C			n_I		n_T	n
Genotype	CC	CI	CT	II	IT	TT	
complete data y	n_{CC}	n_{CI}	n_{CT}	n_{II}	n_{IT}	n_{TT}	n
probability	p_C^2	$2p_Cp_I$	$2p_Cp_T$	p_I^2	$2p_Ip_T$	p_T^2	1

Since $p_T = 1 - p_C - p_I$, the parameter vector is $\theta = (p_C, p_I)$.

$$n_C = n_{CC} + n_{CI} + n_{CT}, \qquad n_I = n_{II} + n_{IT}, \qquad n_T = n_{TT}$$

Example: Log-Likelihood Function

The log-likelihood function, ignoring some constants, is

$$\begin{aligned}\ell(\theta) = n_C \log(p_C^2 + 2p_C p_I + 2p_C p_T) + n_I \log(p_I^2 + 2p_I p_T) \\ + n^T \log(p_T^2)\end{aligned}$$

If we know the genotype, the complete log-likelihood function is

$$\begin{aligned}\ell_c(\theta) &= n_{CC} \log(p_C^2) + n_{CI} \log(2p_C p_I) + n_{CT} \log(2p_C p_T) \\ &\quad + n_{II} \log(p_I^2) + n_{IT} \log(2p_I p_T) + n_{TT} \log(p_T^2) \\ &= (2n_{CC} + n_{CI} + n_{CT}) \log(p_C) + (n_{CI} + 2n_{II} + n_{IT}) \log(p_I) \\ &\quad + (n_{CT} + n_{IT} + 2n_{TT}) \log(p_T^2)\end{aligned}$$

Gaussian Mixture Models

Let $x_1, \ldots, x_n$ be iid random variables sampled from a mixture of K normal distributions. The goal of the analysis is to

- Estimate the parameters $\theta = \{(\pi_k, \mu_k, \sigma_k^2), k = 1, \ldots, K\}$
- Judge from which population one observation is sampled from

Let $\phi(x; \mu, \sigma^2)$ be the density of N(μ, σ^2). The log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \phi(x_i; \mu_k, \sigma_k^2) \right)$$

Let $z_i \in \{1, \ldots, K\}$ be the population from which x_i is sampled from. The complete log-likelihood function is

$$\ell_c(\theta) = \sum_{i=1}^{n} \left\{ \log(\pi_{z_i}) + \log \phi(x_i; \mu_{z_i}, \sigma_{z_i}^2) \right\}$$

Missing Data

- x: observed data with density $f_X(x \mid \theta)$
- z: missing data
- y: complete data with density $f_Y(y \mid \theta)$, assume $x = m(y)$
- θ: parameters to be estimated

We have

$$f_X(x \mid \theta) = \int_{y:m(y)=x} f_Y(y \mid \theta) dy, \quad f_{Z|X}(z \mid x, \theta) = \frac{f_Y(y \mid \theta)}{f_X(x \mid \theta)}$$

In many scenarios, $y = (x, z)$ and we have

$$f_X(x \mid \theta) = E_Z[f_{(X,Z)}(x, Z)], \quad f_{Z|X}(z \mid x, \theta) = \frac{f_{(X,Z)}(x, z \mid \theta)}{f_X(x \mid \theta)}$$

EM Algorithm

Let $\ell(\theta \mid x)$ be the log-likelihood. The goal is to find the MLE of θ.

$$\max_{\theta} \ell(\theta \mid x)$$

The EM is initiated from $\theta^{(0)}$ then alternates between two steps until a stopping rule has been met.

- E-step: compute $Q(\theta \mid \theta^{(t)})$, the expectation of the joint log-likelihood for the complete data, conditional on $X = x$,

$$Q(\theta \mid \theta^{(t)}) = E\Big\{\ell_c(\theta \mid Y) \mid x, \theta^{(t)}\Big\}$$

- M-step: maximize $Q(\theta \mid \theta^{(t)})$ with respect to θ. Set $\theta^{(t+1)}$ as the maximizer of Q

Example: Multinomial Model - E Step

The complete data log-likelihood function is, ignoring some constant,

$$\begin{aligned}\ell_c(\theta) = &(2n_{CC} + n_{CI} + n_{CT}) \log(p_C) + (n_{CI} + 2n_{II} + n_{IT}) \log(p_I) \\ &+ (n_{CT} + n_{IT} + 2n_{TT}) \log(p_T)\end{aligned}$$

Conditional on $\theta^{(t)}$ and n_C, the latent count (n_{CC}, n_{CI}, n_{CT}) follows multinomial distribution with probabilities $(p_C^{(t)})^2$, $2p_C^{(t)} p_I^{(t)}$, $2p_C^{(t)} p_T^{(t)}$.

$$E(N_{CC} \mid n_C, n_I, n_T, \theta^{(t)}) = n_{CC}^{(t)} = \frac{n_C (p_C^{(t)})^2}{(p_C^{(t)})^2 + 2p_C^{(t)} p_I^{(t)} + 2p_C^{(t)} p_T^{(t)}}$$

Evaluate the other conditional expectation similarly. Hence,

$$\begin{aligned}Q(\theta \mid \theta^{(t)}) = &(2n_{CC}^{(t)} + n_{CI}^{(t)} + n_{CT}^{(t)}) \log(p_C) \\ &+ (n_{CI}^{(t)} + 2n_{II}^{(t)} + n_{IT}^{(t)}) \log(p_I) + (n_{CT}^{(t)} + n_{IT}^{(t)} + 2n_{TT}^{(t)}) \log(p_T)\end{aligned}$$

Example: Multinomial Model - M Step

Maximize $Q(\theta \mid \theta^{(t)})$ to obtain $\theta^{(t+1)}$

$$p_C^{(t+1)} = \frac{2n_{CC}^{(t)} + n_{CI}^{(t)} + n_{CT}^{(t)}}{2n}$$

$$p_I^{(t+1)} = \frac{n_{CI}^{(t)} + 2n_{II}^{(t)} + n_{IT}^{(t)}}{2n}$$

$$p_T^{(t+1)} = \frac{n_{CT}^{(t)} + n_{IT}^{(t)} + 2n_{TT}^{(t)}}{2n}$$

Start from an initial value $\{n_{CC}^{(0)}, n_{CI}^{(0)}, n_{CT}^{(0)}, n_{II}^{(0)}, n_{IT}^{(0)}, n_{TT}^{(0)}\}$, and keep updating $\{p_C^{(t+1)}, p_I^{(t+1)}, p_T^{(t+1)}\}$ until convergence.

The observed data are $n_C = 85$, $n_I = 196$, and $n_T = 341$. The MLE is $\hat{p}_C = 0.07084$, $\hat{p}_I = 0.18874$, and $\hat{p}_T = 0.74043$.

Convergence of EM Algorithm

Each M-step increases the observed-data log likelihood $\ell(\theta \mid x)$.

Taking expectation with respect to $Z \mid (x, \theta^{(t)})$ yields

$$\begin{aligned}\log f_X(x \mid \theta) &= \log f_Y(y \mid \theta) - \log f_{Z|X}(z \mid x, \theta) \\ &= Q(\theta \mid \theta^{(t)}) - H(\theta \mid \theta^{(t)})\end{aligned}$$

where

$$H(\theta \mid \theta^{(t)}) = E[\log f_{Z|X}(z \mid x, \theta) \mid x, \theta^{(t)}]$$

Note $\theta^{(t+1)}$ maximize $Q(\theta \mid \theta^{(t)})$. If $H(\theta^{(t)} \mid \theta^{(t)}) - H(\theta \mid \theta^{(t)}) \geq 0$,

$$\log f_X(x \mid \theta^{(t+1)}) - \log f_X(x \mid \theta^{(t)}) \geq 0$$

Convergence of EM Algorithm, Continued

In fact, $H(\theta \mid \theta^{(t)})$ is maximized at $\theta = \theta^{(t)}$. For any θ,

$$
\begin{aligned}
& H(\theta^{(t)} \mid \theta^{(t)}) - H(\theta \mid \theta^{(t)}) \\
& = E\left\{\log f_{Z|X}(z \mid x, \theta^{(t)}) - \log f_{Z|X}(z \mid x, \theta) \mid x, \theta^{(t)}\right\} \\
& = \int -\log\left[\frac{f_{Z|X}(z \mid x, \theta)}{f_{Z|X}(z \mid x, \theta^{(t)})}\right] f_{Z|X}(z \mid x, \theta^{(t)}) dz \\
& \geq -\log \int \left[\frac{f_{Z|X}(z \mid x, \theta)}{f_{Z|X}(z \mid x, \theta^{(t)})}\right] f_{Z|X}(z \mid x, \theta^{(t)}) dz \\
& = 0
\end{aligned}
$$

where Jensen's inequality is applied to function $-\log(u)$.

$$E[-\log(U)] \geq -\log[E(U)]$$

Comments

- A generalized EM algorithm simply selects $\theta^{(t+1)}$ such that $Q(\theta^{(t+1)} \mid \theta^{(t)}) > Q(\theta^{(t)} \mid \theta^{(t)})$. where $\theta^{(t+1)}$ does not max Q.
- The EM algorithm has linear convergence when $p = 1$. When $p > 1$, it is linear provided that the observed information $-\ell''(\hat{\theta} \mid x)$ is positive definite.
- EM suffers slower convergence when the proportion of missing information is larger. Let ρ be the largest eigenvalue of the Jacobian of the mapping $\theta^{(t+1)} = \Psi(\theta^{(t)})$. It can be shown that

$$\rho = \lim_{t \to \infty} \frac{\|\theta^{(t+1)} - \hat{\theta}\|}{\|\theta^{(t)} - \hat{\theta}\|}$$

  A larger ρ indicates slower convergence and more missing.

Exponential Family

When the complete data have a distribution in the exponential family,

$$f(y \mid \theta) = c_1(y)c_2(\theta)\exp\{\theta^T s(y)\},$$

where θ is the natural parameter and $s(y)$ is sufficient statistics. Let $s^{(t)} = E[s(Y) \mid x, \theta^{(t)}]$. The E-step finds

$$Q(\theta \mid \theta^{(t)}) = \log c_2(\theta) + \theta^T s^{(t)},$$

In the M-step, setting the gradient of $Q(\theta \mid \theta^{(t)})$ to zero yields,

$$s^{(t)} = -(\log c_2(\theta))' = -c_2'(\theta)/c_2(\theta) = E[s(Y) \mid \theta]$$

where the last equality holds because of a property of the exponential family. Hence, we find $\theta^{(t+1)}$ as the root of this nonlinear equation.

EM-Algorithm for Exponential Family

The EM algorithm for exponential families consists of

- E-step: compute the expected values of the sufficient statistics for the complete data, given the observed data and using the current parameter guesses $\theta^{(t)}$. Let

$$s^{(t)} = E[s(Y) \mid x, \theta^{(t)}] = \int s(y) f_{Z|X}(z \mid x, \theta^{(t)}) dz$$

- M-step: set $\theta^{(t+1)}$ to the value that makes the unconditional expectation of the sufficient statistics for the complete data equal to $s^{(t)}$. In another words,

$$\text{find } \theta^{(t+1)} \text{ that solves } E[s(Y) \mid \theta] = s^{(t)}$$

Example: Peppered Moths

For a multinomial distribution,

$$f(y) = \frac{n!}{y_1! \cdots y_k!} p_1^{y_1} \cdots p_k^{y_k} = \frac{n!}{y_1! \cdots y_k!} \exp\left\{ \sum_{i=1}^{k} y_i \log p_i \right\}$$

Hence,

- Sufficient statistics: genotype counts n_{CC}, n_{CI}, n_{CT}, ...
- Natural parameters: log probabilities

Apply the results in the previous slide,

- E-step: the sufficient statistics are

$$s_{CC}^{(t)} = n_{CC}^{(t)}, \quad s_{CI}^{(t)} = n_{CI}^{(t)}, \quad s_{CT}^{(t)} = n_{CT}^{(t)}, \quad \ldots$$

- M-step: solve the equations to get the same updating equations

$$np_C^2 = n_{CC}^{(t)}, \quad 2np_C p_I = n_{CI}^{(t)}, \quad 2np_C p_T = n_{CT}^{(t)}, \quad \ldots$$

Variance Estimation

Let the MLE be $\hat{\theta}$. The variance of $\hat{\theta}$ is

$$var(\hat{\theta}) = [I(\theta)]^{-1} = [-\ell''(\theta \mid x)]^{-1}$$

where ℓ'' is the Hessian matrix of log-likelihood.

If it is difficult to directly compute $-\ell''(\hat{\theta} \mid x)$ analytically,

- Louis's method
- Supplemented EM algorithm
- Bootstrapping
- Numerical differentiation

Observed Information

Recall that

$$\log f_X(x \mid \theta) = Q(\theta \mid \theta^{(t)}) - H(\theta \mid \theta^{(t)})$$

where $H(\theta \mid \theta^{(t)}) = E[\log f_{Z|X}(Z \mid x, \theta) \mid x, \theta^{(t)}]$. Hence,

$$-\ell''(\theta \mid x) = -Q''(\theta \mid w)\Big|_{w=\theta^{(t)}} + H''(\theta \mid w)\Big|_{w=\theta^{(t)}}$$

where Q'' and H'' are derivatives with respect to the first argument.

- $\hat{I}_X(\theta) = -\ell''(\theta \mid x)$ observed information
- $\hat{I}_Y(\theta) = -Q''(\theta \mid w)\Big|_{w=\theta}$ complete information
- $\hat{I}_{Z|X}(\theta) = -H''(\theta \mid w)\Big|_{w=\theta}$ missing information

Missing-Information Principle

Interchanging integration and differentiation (if possible), we have

$$\hat{I}_Y(\theta) = -Q''(\theta \mid w)\Big|_{w=\theta} = -E[\ell_c''(\theta \mid Y) \mid x, \theta]$$

Similarly,

$$\hat{I}_{Z|X}(\theta) = var\left\{\frac{\partial \log f_{Z|X}(Z \mid x, \theta)}{\partial \theta}\right\}$$

The observed information equals the complete information minus the missing information.

$$\hat{I}_X(\theta) = \hat{I}_Y(\theta) - \hat{I}_{Z|X}(\theta)$$

Louis's Method

In many scenarios, $\hat{I}_Y(\theta)$ and $\hat{I}_{Z|X}(\theta)$ are easier to evaluate. If they are difficult to compute, we can use the Monte Carlo method.

$$\hat{I}_Y(\theta) \approx -\frac{1}{m}\sum_{i=1}^{m} \frac{\partial^2 \log f_Y(y_i \mid \theta)}{\partial\theta\partial\theta^T}$$

where y_i are simulated complete datasets consisting of the observed data and iid imputed missing data z_i drawn from $f_{Z|X}$. Similarly,

$$\hat{I}_{Z|X}(\theta) \approx \text{sample variance of } \frac{\partial \log f_{Z|X}(z_i \mid x, \theta)}{\partial\theta}$$

Example: Censored Exponential Data

Let the complete data be $Y_1, \ldots, Y_n \sim \text{exponential}(\lambda)$. Some cases are right-censored and the observed data are $x_i = (\min(y_i, c_i), \delta_i)$, where c_i is the censoring levels, $\delta_i = 1$ if $y_i \leq c_i$ and $\delta_i = 0$ otherwise.

The complete-data log-likelihood is

$$\ell(\lambda) = n \log \lambda - \lambda \sum_i y_i$$

The unobserved outcome Z_i for a censored case has density

$$f_{Z_i|X}(z_i \mid x, \lambda) = \lambda \exp\{-\lambda(z_i - c_i)\}, \quad z_i > c_i$$

Therefore, when $\delta_i = 0$, $E(Y_i \mid x_i) = E(Y_i \mid Y_i > c_i) = c_i + \lambda^{-1}$.
When $\delta_i = 1$, $E(Y_i \mid x_i) = y_i$.

Example, Continued

Let $C = \sum_i (1-\delta_i)$ be the number of censored cases, and $U = \sum_i \delta_i$ be the number of uncensored cases.

$$Q(\lambda \mid \lambda^{(t)}) = n \log \lambda - \lambda \sum_{i=1}^{n} [y_i \delta_i + c_i(1-\delta_i)] - \frac{C\lambda}{\lambda^{(t)}}$$

Hence, $-Q''(\lambda \mid \lambda^{(t)}) = n/\lambda^2$. Similarly,

$$\hat{I}_{Z|X}(\lambda) = \frac{C}{\lambda^2}$$

Apply Louis's method,

$$\hat{I}_X(\lambda) = \frac{n}{\lambda^2} - \frac{C}{\lambda^2} = \frac{U}{\lambda^2}$$

Supplemented EM Algorithm

Let Ψ be the EM mapping, that is, $\theta^{(t+1)} = \Psi(\theta^{(t)})$. Then $\hat{\theta}$ is the fixed point of Ψ. Let $\Psi'(\theta)$ be the Jacobian. It can be shown that

$$\Psi'(\hat{\theta})^T = \hat{I}_{Z|X}(\hat{\theta})\hat{I}_Y(\hat{\theta})^{-1}.$$

Hence,

$$\hat{I}_X(\hat{\theta}) = \hat{I}_Y(\hat{\theta}) - \hat{I}_{Z|X}(\hat{\theta}) = [I - \Psi'(\hat{\theta})^T]\hat{I}_Y(\hat{\theta})$$

We can have

$$\begin{aligned} \widehat{var}(\hat{\theta}) &= \hat{I}_Y(\hat{\theta})^{-1}[I - \Psi'(\hat{\theta})^T]^{-1} \\ &= \hat{I}_Y(\hat{\theta})^{-1} + \hat{I}_Y(\hat{\theta})^{-1}\Psi'(\hat{\theta})^T[I - \Psi'(\hat{\theta})^T]^{-1} \end{aligned}$$

which equals the complete data covariance plus an incremental matrix that takes account of the uncertainty due to the missing data.

SEM Algorithm

The supplemented EM (SEM) algorithm estimates the variance using the above formula, and compute Ψ' using numerical differentiation.

- Run the EM algorithm to convergence to find the maximizer $\hat{\theta}$
- Restart the algorithm from $\theta^{(0)}$, which is commonly selected to be close to $\hat{\theta}$. For $t = 0, 1, 2, \ldots$,
  - Take a standard E-step and M-step to produce $\theta^{(t+1)}$ from $\theta^{(t)}$
  - For $j = 1, \ldots, p$, define $\theta^{(t)}(j) = (\ldots, \hat{\theta}_{j-1}, \theta_j^{(t)}, \hat{\theta}_{j+1}, \ldots)$ and

$$r_{ij}^{(t)} = \frac{\Psi(\theta^{(t)}(j)) - \hat{\theta}_j}{\theta_j^{(t)} - \hat{\theta}_j},$$

which converges to $(\Psi'(\hat{\theta}))_{ij}$ as $t \to \infty$.

Comments

- Different numbers of iterations may be needed for precise estimation of different elements of $\Psi'(\hat{\theta})$. When all elements have stabilized, SEM iterations stop and the resulting estimate of $\Psi'(\hat{\theta})$ is used to determine $\widehat{var}(\hat{\theta})$.
- Numerical imprecision can cause the resulting covariance matrix to be slightly asymmetric. Such asymmetry can be used to diagnose whether the original EM procedure was run to sufficient precision and to assess how many digits are trustworthy in entries of the estimated covariance matrix.
- Transforming θ to achieve an approximately normal likelihood can lead to faster convergence and increased accuracy of the final solution.

Results of Peppered Moths

The MLE is $\hat{p}_C = 0.07084$, $\hat{p}_I = 0.18874$, and $\hat{p}_T = 0.74043$.

Restart from $p_C^{(0)} = 0.07$, $p_I^{(0)} = 0.19$, and $p_T^{(0)} = 0.74$.

The standard errors are

$$\mathsf{SE}(\hat{p}_C) = 0.0074, \quad \mathsf{SE}(\hat{p}_I) = 0.0119, \quad \mathsf{SE}(\hat{p}_I) = 0.0132$$

Pairwise correlations are

$$\mathsf{cor}(\hat{p}_C, \hat{p}_I) = -0.14, \quad \mathsf{cor}(\hat{p}_C, \hat{p}_T) = -0.44, \quad \mathsf{cor}(\hat{p}_I, \hat{p}_T) = -0.83$$

Bootstrapping

- Calculate $\hat{\theta}_{\text{EM}}$ from $x_1, \ldots, x_n$.
- For $j = 1, \ldots, B$,
  - Sample pseudo-data $x_1^*, \ldots, x_n^*$ completely at random from $x_1, \ldots, x_n$ with replacement.
  - Calculate $\hat{\theta}_j$ from $x_1^*, \ldots, x_n^*$ using the same approach

For most problems, a few thousand iterations will suffice.

$$var(\hat{\theta}_{\text{EM}}) = \text{sample variance of } \hat{\theta}_1, \ldots, \hat{\theta}_B$$

Similarly, other aspects of the sampling distribution of $\hat{\theta}$, such as correlations and quantiles, can be estimated using the corresponding sample estimates based on $\hat{\theta}_1, \ldots, \hat{\theta}_B$.

Empirical Information

The score function

$$\frac{\partial \log f_X(x \mid \theta)}{\partial \theta} = \ell'(\theta \mid x) = \sum_{i=1}^{n} \ell'(\theta \mid x_i)$$

The Fisher information is the variance of the score function, hence can be estimated by the empirical information, which is the sample variance of the individual scores,

$$\frac{1}{n}\sum_{i=1}^{n} \ell'(\theta \mid x_i)\ell'(\theta \mid x_i)^T - \frac{1}{n^2}\ell'(\theta \mid x)\theta'(\theta \mid x)^T$$

Because $\theta^{(t)}$ maximizes $H(\theta \mid \theta^{(t)}) = Q(\theta \mid \theta^{(t)}) - \ell(\theta \mid x)$, (slide 13)

$$Q'(\theta \mid \theta^{(t)})\Big|_{\theta=\theta^{(t)}} = \ell'(\theta \mid x)\Big|_{\theta=\theta^{(t)}}$$

Since Q' is usually calculated at each M-step, the empirical information can be easily computed.

Numerical Differentiation

To estimate the Hessian, we can compute the numerical derivative of ℓ' at $\hat{\theta}$, one coordinate at a time.

$$H_{ij} = \frac{\ell_i'(\hat{\theta} + \delta_{ij} e_j) - \ell'(\hat{\theta})}{\delta_{ij}}$$

If a perturbation δ_{ij} is too small, estimated partial derivatives may be inaccurate due to roundoff error; if a perturbation δ_{ij} is too big, the estimates may also be inaccurate.

Improving the E-Step: MCEM

Monte Carlo EM (MCEM) algorithm can be used if it is difficult to find $Q(\theta \mid \theta^{(t)})$ explicitly in the E-step.

- Draw missing datasets $z_1^{(t)}, \ldots, z_{m^{(t)}}^{(t)}$ iid from $f_{Z|X}(z \mid x, \theta^{(t)})$, where each $z_j^{(t)}$ is a vector of all the missing values needed to complete the observed dataset.
- Let $y_j = (x, z_j)$ with missing values replaced by z_j and calculate

$$\hat{Q}^{(t+1)}(\theta \mid \theta^{(t)}) = \frac{1}{m^{(t)}} \sum_{j=1}^{m^{(t)}} \log f_Y(y_j^{(t)} \mid \theta).$$

  Then $\hat{Q}^{(t+1)}(\theta \mid \theta^{(t)})$ is a Monte Carlo estimate of $Q(\theta \mid \theta^{(t)})$.

It is recommended to let $m^{(t)}$ be small during early EM iterations and to increase $m^{(t)}$ as iterations progress to reduce the Monte Carlo variability introduced in $\hat{Q}$.

Example: Censored Exponential Data

Recall that

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^{n} y_i$$

We know that $Z_i - c_i \sim \text{exponential}(\lambda)$. Sample z_{ij} independently from $c_i + \text{exponential}(\lambda^{(t)})$ for $i = 1, \ldots, n$ and $j = 1, \ldots, m^{(t)}$. Then

$$\hat{Q}^{(t+1)}(\lambda \mid \lambda^{(t)}) \approx n \log \lambda - \frac{\lambda}{m^{(t)}} \sum_{j=1}^{m^{(t)}} \sum_{i=1}^{n} y_{ij}$$

where $y_{ij} = x_i$ if $\delta_i = 1$ and $y_{ij} = z_{ij}$ if $\delta_i = 0$.

$$\hat{\lambda}^{(t+1)} = \frac{n}{\sum_{j=1}^{m^{(t)}} \sum_{i=1}^{n} y_{ij} / m^{(t)}}$$

Improving the M-Step

The ECM algorithm replaces the M step with a series of computationally simpler conditional maximization (CM) steps.

- CM cycle: collection of CM steps after one E step
- $S =$ the total number of CM steps in one CM cycle.
- For $s = 1, \ldots, S$, the sth CM step in the tth cycle requires the maximization of $Q(\theta \mid \theta^{(t)})$ subject to a constraint, say

$$g_s(\theta) = g_s(\theta^{(t+(s-1)/S)})$$

  where $\theta^{(t+(s-1)/S)}$ is the maximizer found in the $(s-1)$th CM step of the current cycle.
- When the entire cycle of S steps of CM has been completed, set $\theta^{(t+1)} = \theta^{(t+S/S)}$ and proceed to the E-step in the next iteration.

Comments

- An ECM is a GEM algorithm, since each CM step increases Q.
- In order for ECM to converge, we need to ensure that each CM cycle permits search in any direction for a maximizer of $Q(\theta \mid \theta(t))$, so that ECM effectively maximizes over the original parameter space for θ and not over some subspace.
- One possible way to select g_s: partition θ into S subvectors, $\theta = (\theta_1, \ldots, \theta_S)$. Then in the sth CM step, one might seek to maximize Q with respect to θ_s while holding all other components of θ fixed.
- A variant of ECM inserts an E step between each pair of CM steps, thereby updating Q at every stage of the CM cycle.

Multivariate Regression with Missing Values

Let $U_i \in \mathbb{R}^d$ be an iid sample from $U_i \sim N_d(\mu_i, \Sigma)$ with $\mu_i = V_i\beta$. Some elements of some U_i are missing.

$$U_i = (U_{\text{obs},i}, U_{\text{miss},i}),\ \mu_i = (\mu_{\text{obs},i}, \mu_{\text{miss},i}),\ \Sigma_i = \begin{pmatrix} \Sigma_{\text{obs},i} & \Sigma_{\text{cross},i} \\ \Sigma_{\text{cross},i} & \Sigma_{\text{miss},i} \end{pmatrix}$$

The full set of observed data is $U_{\text{obs}} = (U_{\text{obs},1}, \ldots, U_{\text{obs},n})$. The observed-data log-likelihood is

$$\ell(\beta, \Sigma) = \frac{1}{2}\sum_{i=1}^n \log|\Sigma_{\text{obs},i}| - \frac{1}{2}\sum_{i=1}^n (U_{\text{obs},i} - \mu_{\text{obs},i})^T \Sigma_{\text{obs},i}^{-1}(\cdots)$$

The complete-data sufficient statistics are given by $\sum_i U_{ij}$ and $\sum_i U_{ij}U_{ik}$ for $j, k = 1, \ldots, d$.

Example: E-Step

E-step: find the expectation of these sufficient statistics conditional on the observed data and $\beta^{(t)}$ and $\Sigma^{(t)}$.

For $j = 1, \ldots, d$,

$$E\left\{\sum_{i=1}^{n} U_{ij} \mid u_{\text{obs}}, \beta^{(t)}, \Sigma^{(t)}\right\} = \sum_{i=1}^{n} a_{ij}^{(t)},$$

where $\alpha_{ij}^{(t)} = E(U_{ij} \mid u_{\text{obs},i}, \beta_i^{(t)}, \Sigma_i^{(t)})$ and

$$a_{ij}^{(t)} = \begin{cases} \alpha_{ij}^{(t)}, & \text{if } U_{ij} \text{ is missing} \\ u_{ij}, & \text{if } U_{ij} \text{ is observed} \end{cases}$$

For $j, k = 1, \ldots, d$,

$$E\left\{\sum_{i=1}^{n} U_{ij}U_{ik} \mid u_{\text{obs}}, \beta^{(t)}, \Sigma^{(t)}\right\} = \sum_{i=1}^{n}(a_{ij}^{(t)}a_{ik}^{(t)} + b_{ijk}^{(t)})$$

where $\gamma_{ijk}^{(t)} = cov(U_{ij}, U_{ik} \mid u_{\text{obs},i}, \beta_i^{(t)}, \Sigma_i^{(t)})$ and

$$b_{ijk}^{(t)} = \begin{cases} \gamma_{ijk}^{(t)}, & \text{if } U_{ij} \text{ and } U_{ik} \text{ are both missing} \\ 0, & \text{otherwise} \end{cases}$$

Note that $\alpha_{ij}^{(t)}$ and $\gamma_{ijk}^{(t)}$ can be obtained from the conditional distribution of $U_{\text{miss},i} \mid (u_{\text{obs},i}, \beta_i^{(t)}, \Sigma_i^{(t)})$,

$$N(\mu_{\text{miss},i}^{(t)} + \Sigma_{\text{cross},i}\Sigma_{\text{miss},i}^{-1}(u_{\text{obs},i} - \mu_{\text{obs},i}^{(t)}), \Sigma_{\text{obs},i} - \Sigma_{\text{cross},i}\Sigma_{\text{miss},i}^{-1}\Sigma_{\text{cross},i}^{T})$$

Example: M-Step

Each CM cycle contains two steps: update β, and then update Σ.

- Fix $\Sigma = \Sigma^{(t)}$, update β using weighted least squares,

$$\beta^{(t+1/2)} = \left(\sum_{i=1}^{n} V_i^T (\Sigma_i^{(t)})^{-1} V_i \right)^{-1} \left(\sum_{i=1}^{n} V_i^T (\Sigma_i^{(t)})^{-1} a_i^{(t)} \right)$$

- Fix $\beta = \beta^{(t+1/2)}$,

$$\Sigma^{(t+2/2)} = E \left\{ \frac{1}{n} \sum_{i=1}^{n} (U_i - V_i \beta^{(t+1/2)})(U_i - V_i \beta^{(t+1/2)})^T \right\}$$

We need to plug in $\alpha_{ij}^{(t)}$ and $\gamma_{ijk}^{(t)}$ where necessary.

$$\Sigma_{jk}^{(t+2/2)} = \frac{1}{n} \sum_{i=1}^{n} (U_{ij} U_{ik} - \mu_{ij} U_{ik} - U_{ij} \mu_{ik} + \mu_{ij} \mu_{ik})$$

EM Gradient Algorithm

If maximization cannot be accomplished analytically, then one might consider carrying out each M step using an iterative numerical optimization approach.

Replace the M-step with a single step of Newton's method.

$$\begin{aligned}\theta^{(t+1)} &= \theta^{(t)} - Q''(\theta \mid \theta^{(t)})^{-1}\Big|_{\theta=\theta^{(t)}} Q'(\theta \mid \theta^{(t)})\Big|_{\theta=\theta^{(t)}} \\ &= \theta^{(t)} - Q''(\theta \mid \theta^{(t)})^{-1}\Big|_{\theta=\theta^{(t)}} \ell'(\theta^{(t)} \mid x)\end{aligned}$$

where ℓ' is the score function.

- The second equality holds because $\theta^{(t)}$ maximizes $H(\theta \mid \theta^{(t)}) = Q(\theta \mid \theta^{(t)}) - \ell(\theta \mid x)$. (Slide 13)
- This EM gradient algorithm has the same rate of convergence to $\hat{\theta}$ as the full EM algorithm.

Aitken Acceleration

Let $\theta_{\text{EM}}^{(t+1)}$ be the next iterate obtained by the standard EM algorithm from $\theta^{(t)}$. Recall that the Newton update would be

$$\theta^{(t+1)} = \theta^{(t)} - \ell''(\theta^{(t)} \mid x)^{-1} \ell'(\theta^{(t)} \mid x)$$

Note that $\ell'(\theta^{(t)} \mid x) = Q'(\theta \mid \theta^{(t)}) \mid_{\theta=\theta^{(t)}}$. Expanding Q' around $\theta^{(t)}$, evaluated at $\theta_{\text{EM}}^{(t+1)}$, yields,

$$0 = Q'(\theta \mid \theta^{(t)})\Big|_{\theta=\theta_{\text{EM}}^{(t+1)}} \approx Q'(\theta \mid \theta^{(t)})\Big|_{\theta=\theta^{(t)}} - \hat{I}_Y(\theta^{(t)})(\theta_{\text{EM}}^{(t+1)} - \theta^{(t)})$$

Hence, the updating equation becomes

$$\theta^{(t+1)} = \theta^{(t)} - \ell''(\theta^{(t)} \mid x)^{-1} \hat{I}_Y(\theta^{(t)})(\theta_{\text{EM}}^{(t+1)} - \theta^{(t)})$$

The approach is the same as applying the Newton–Raphson method to find a zero of $\Psi(\theta) - \theta$.

Comments

- Aitken acceleration is sometimes criticized for potential numerical instabilities and convergence failure.
- The approximation only becomes a precise approximation as $\theta^{(t)}$ nears $\hat{\theta}$. We should only expect this acceleration approach to enhance convergence only as preliminary iterations hone θ sufficiently.

Quasi-Newton Acceleration

The quasi-Newton optimization method produces updates by

$$\theta^{(t+1)} = \theta^{(t)} - (M^{(t)})^{-1}\ell'(\theta^{(t)} \mid x)$$

Note that

$$\ell''(\theta^{(t)} \mid x) = Q''(\theta \mid \theta^{(t)})\Big|_{\theta=\theta^{(t)}} - H''(\theta \mid \theta^{(t)})\Big|_{\theta=\theta^{(t)}}$$

We want to approximate the last term by $B^{(t)}$.
Start with $B^{(0)} = 0$ and gradually accumulate information about H'' as iterations progress. We can require $B^{(t)}$ satisfy

$$B^{(t+1)}a^{(t)} = b^{(t)},$$

where $a^{(t)} = \theta^{(t+1)} - \theta^{(t)}$ and

$$b^{(t)} = H'(\theta \mid \theta^{(t+1)})\Big|_{\theta=\theta^{(t+1)}} - H'(\theta \mid \theta^{(t+1)})\Big|_{\theta=\theta^{(t)}}$$

We can select

$$B^{(t+1)} = B^{(t)} + c^{(t)} v^{(t)} (v^{(t)})^T$$

where $v^{(t)} = b^{(t)} - B^{(t)} a^{(t)}$ and $c^{(t)} = 1/[(v^{(t)})^T a^{(t)}]$.

- Note that $b^{(t)}$ may be expressed entirely in terms of Q'.

$$b^{(t)} = Q'(\theta \mid \theta^{(t)})\Big|_{\theta=\theta^{(t)}} - Q'(\theta \mid \theta^{(t+1)})\Big|_{\theta=\theta^{(t)}}$$

- If $(v^{(t)})^T a^{(t)} = 0$ or is too small, we may set $B^{(t+1)} = B^{(t)}$.
- If $M^{(t)}$ is negative definite, set $M^{(t)} = Q''(\theta^{(t)} \mid \theta^{(t)}) - \alpha^{(t)} B^{(t)}$. where $\alpha^{(t)} = 2^{-m}$ is chosen to make $M^{(t)}$ positive definite.